

Comparison to the 2 layer neural network

The DistilBERT model can use raw text (program names, comments), not just six numeric features like the Module 12 network. That extra signal helped as the accuracy went from about 69 percent to 79 percent, with an F1 of 0.753. Comments and program text do seem to help, especially since GRE is missing for most applicants, so the model leans on text and GPA more than test scores. The pretrained model is more flexible since it picks up on phrasing and context the numeric-only model never saw, but it's also more fragile in practice: bigger dependencies, needs a GPU to train in reasonable time, and harder to inspect than a simple two-layer network. The accuracy jump makes the added complexity worth it, though the gain is incremental rather than huge, so there's probably a ceiling on this dataset no matter which model you use.

Limitations and Ethics

Fine-tuning a pretrained model instead of training from scratch means it already understands English well, so it needs way less data and time to learn this specific task. Combining text and structured features make sense because real admissions aren't purely numeric either; comments about research experience often carry real signal that GPA and GRE alone miss. That said, self-reported Grad Cafe data has clear bias. People who post here are self-selected and often more anxious or active online than typical applicants, and outcomes are self-reported with zero verification. Popular programs are probably overrepresented too. This model can be misleading because it only sees what people chose to type into a public forum, not the actual factors admissions committees use, like letters of recommendation or interview performance. Stuff like applicant pool size or funding availability that year is completely missing from the dataset. A model like this should never be used for real decisions. It's trained on an unverified, biased sample, and even 79 percent accuracy means it's wrong roughly one in five times, with no way for someone to know if they're in that group. On a public page, uncertainty needs to be handled honestly, showing a confidence score plus a clear disclaimer, not presenting the output like it means anything official. Still, a project like this is genuinely useful as a learning exercise. It walks through the full pipeline of turning messy text and numeric data into a fine-tuned model deployed behind a real web form, which is basically what real ML engineering work looks like. The value is in building and understanding that pipeline, not in the predictions being trustworthy, and that's worth being upfront about to anyone using the page.